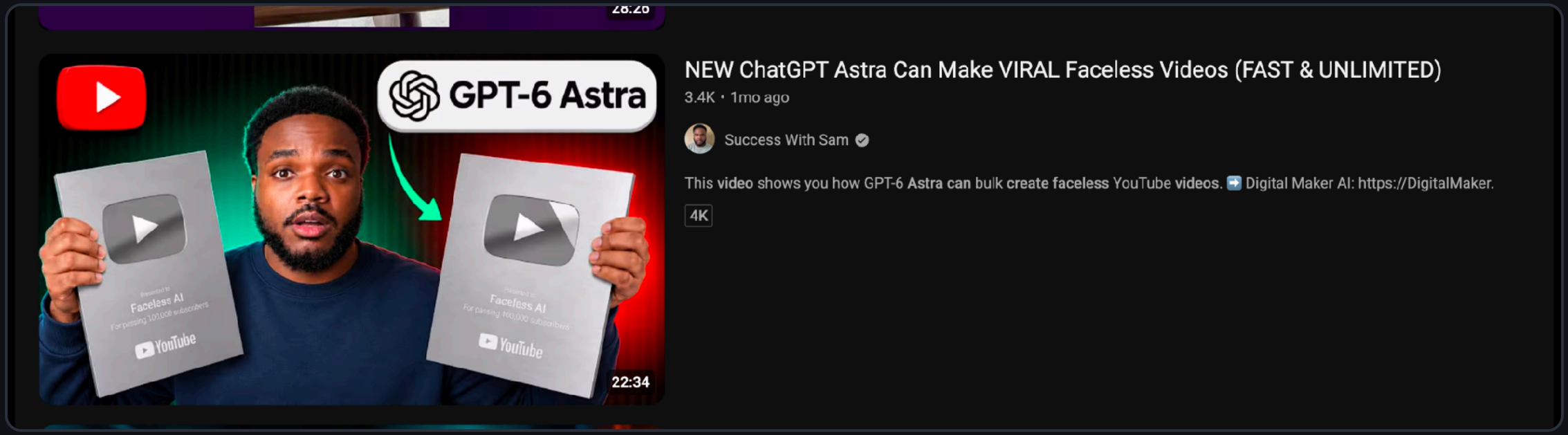
HACKATHON PROJECT

Can RSI improve an LLM's understanding of physics?

BACKGROUND

Advanced LLMs can now generate videos that look physically real.

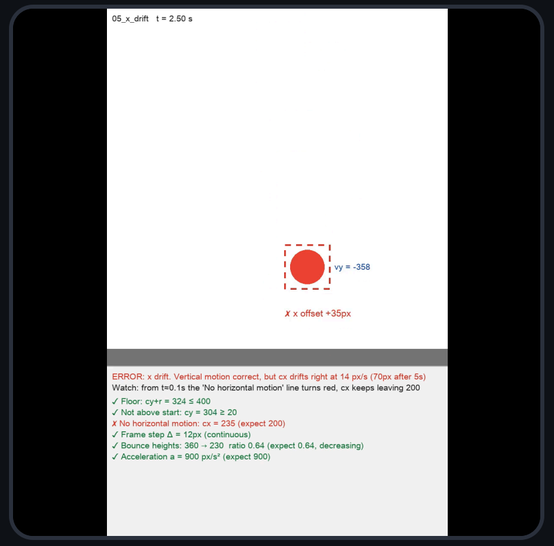


THE PROBLEM

But the motion often breaks the physics.

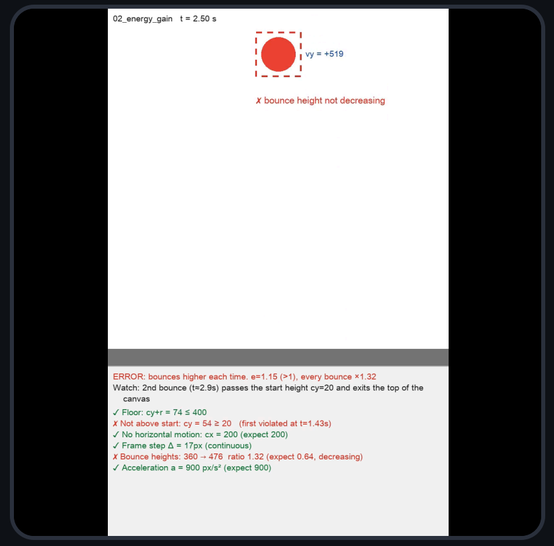
Newton's laws

A ball drifts sideways with no force.



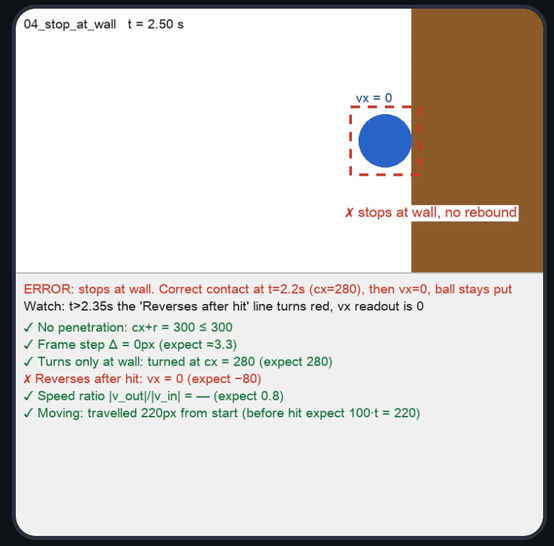
Energy conservation

Each bounce goes higher than the last.



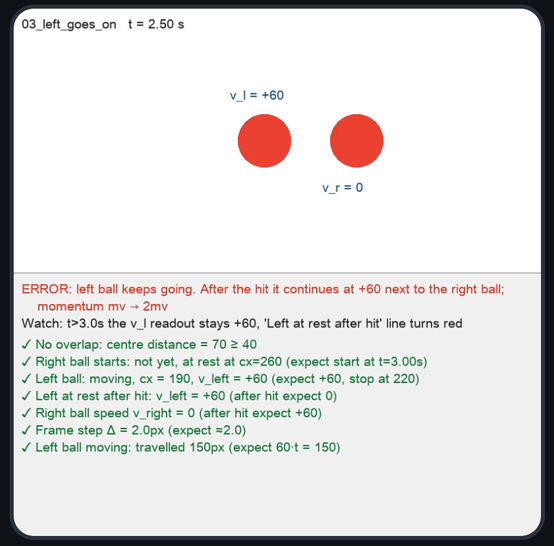
Contact

The ball stops at the wall and does not come back.



Momentum

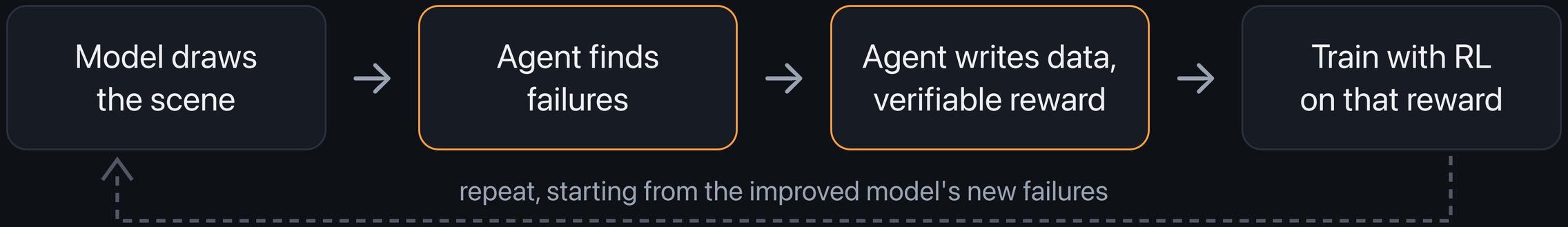
The hitting ball keeps going after a collision.



Example violations, each flagged by a check.

OUR IDEA

Put an agent in the improvement loop.



- The agent, a stronger model (Claude), builds the training environment.
- It generates the data and writes a verifiable reward: a program checks the physics.
- It fine-tunes the model with that reward as the RL signal, and keeps exploring.

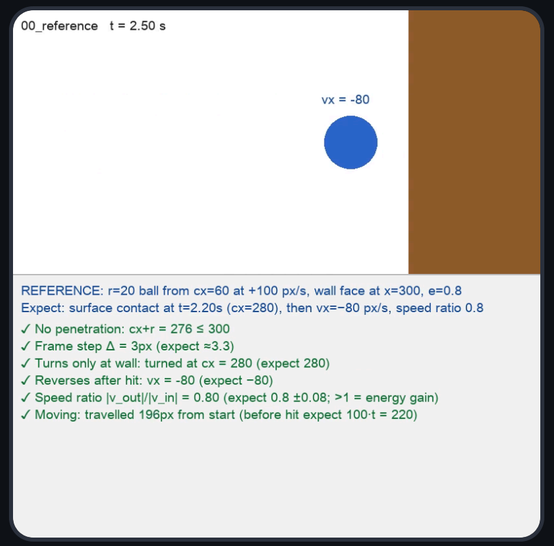
MODEL AND TRAINING

Small model, reinforcement learning.

MODEL	**Small Qwen instruct model** Qwen2.5-0.5B-Instruct
ALGORITHM	**GRPO with LoRA** Hugging Face TRL
REWARD	**Verifiable reward from a checker** A program scores the physics. Written and revised by the agent
DATA	**Randomized ball and wall scenes** Generated by the agent

EXAMPLES

What correct physics looks like.

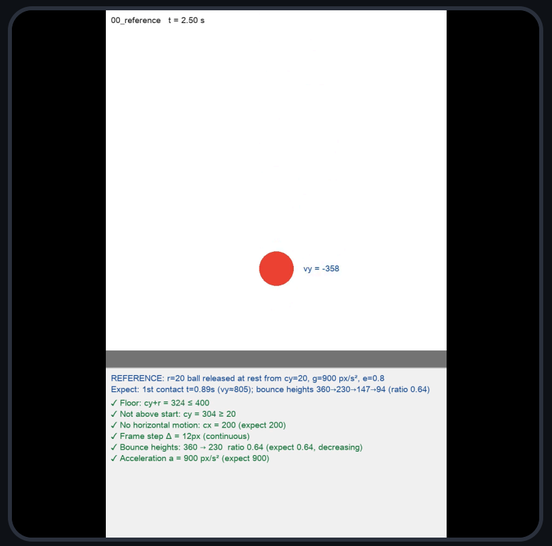


REFERENCE: r=20 ball from cx=60 at +100 px/s, wall face at x=300, e=0.8
Expect: surface contact at t=2.20s (cx=280), then vx=−80 px/s, speed ratio 0.8

✓ No penetration: cx+r = 276 ≤ 300
✓ Frame step Δ = 3px (expect ≈3.3)
✓ Turns only at wall: turned at cx = 280 (expect 280)
✓ Reverses after hit: vx = -80 (expect −80)
✓ Speed ratio |v_out|/|v_in| = 0.80 (expect 0.8 ±0.08; >1 = energy gain)
✓ Moving: travelled 196px from start (before hit expect 100·t = 220)

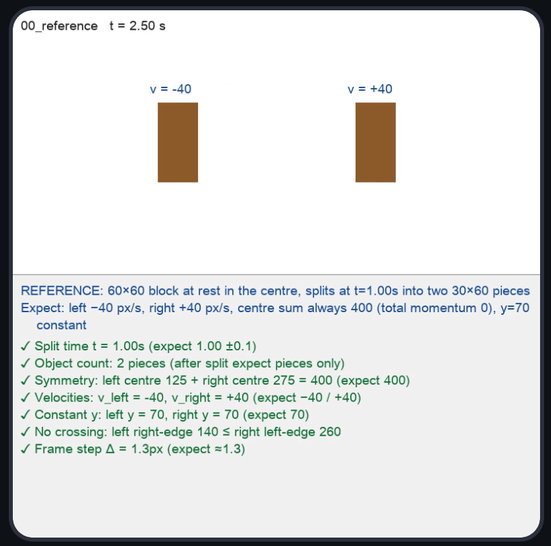
Ball hits a wall

Rebounds at 0.8× its speed



Ball bounces

Bounce heights decay

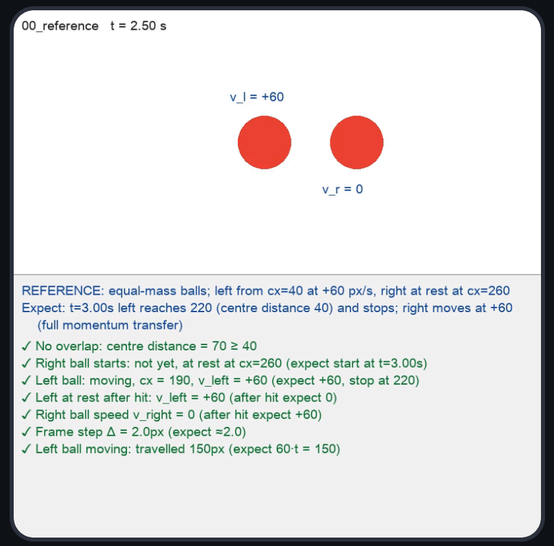


REFERENCE: 60×60 block at rest in the centre, splits at t=1.00s into two 30×60 pieces
Expect: left −40 px/s, right +40 px/s, centre sum always 400 (total momentum 0), y=70 constant

✓ Split time t = 1.00s (expect 1.00 ±0.1)
✓ Object count: 2 pieces (after split expect pieces only)
✓ Symmetry: left centre 125 + right centre 275 = 400 (expect 400)
✓ Velocities: v_left = -40, v_right = +40 (expect −40 / +40)
✓ Constant y: left y = 70, right y = 70 (expect 70)
✓ No crossing: left right-edge 140 ≤ right left-edge 260
✓ Frame step Δ = 1.3px (expect ≈1.3)

Block splits

Momentum stays zero



REFERENCE: equal-mass balls; left from cx=40 at +60 px/s, right at rest at cx=260
Expect: t=3.00s left reaches 220 (centre distance 40) and stops; right moves at +60 (full momentum transfer)

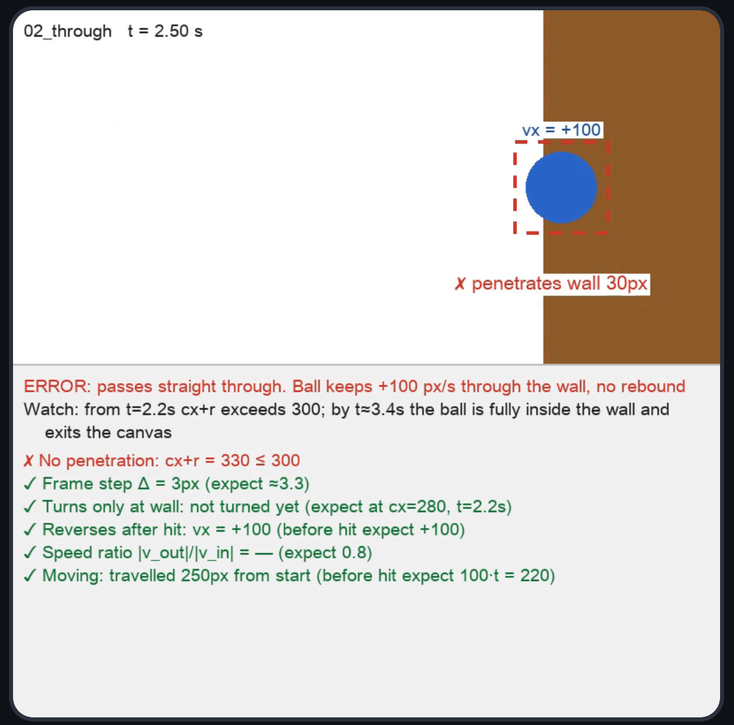
✓ No overlap: centre distance = 70 ≥ 40
✓ Right ball starts: not yet, at rest at cx=260 (expect start at t=3.00s)
✓ Left ball: moving, cx = 190, v_left = +60 (expect +60, stop at 220)
✓ Left at rest after hit: v_left = +60 (after hit expect 0)
✓ Right ball speed v_right = 0 (after hit expect +60)
✓ Frame step Δ = 2.0px (expect ≈2.0)
✓ Left ball moving: travelled 150px (expect 60·t = 150)

Two balls collide

Momentum passes on

FAILURE MODES

Each failure becomes a reward signal.

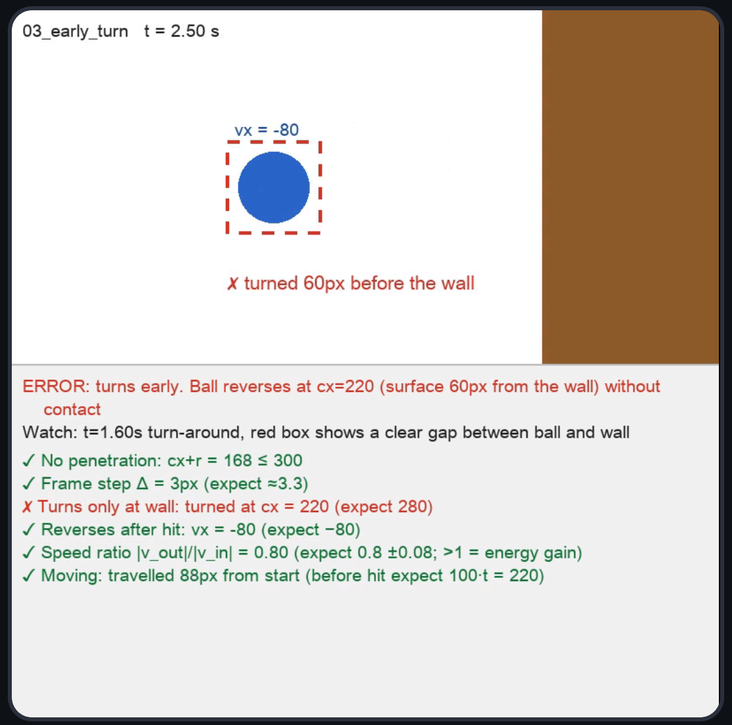


ERROR: passes straight through. Ball keeps +100 px/s through the wall, no rebound
Watch: from t=2.2s cx+r exceeds 300; by t≈3.4s the ball is fully inside the wall and exits the canvas

✗ No penetration: cx+r = 330 ≤ 300
✓ Frame step Δ = 3px (expect ≈3.3)
✓ Turns only at wall: not turned yet (expect at cx=280, t=2.2s)
✓ Reverses after hit: vx = +100 (before hit expect +100)
✓ Speed ratio |v_out|/|v_in| = — (expect 0.8)
✓ Moving: travelled 250px from start (before hit expect 100·t = 220)

Passes through the wall

→ penetration is a hard violation

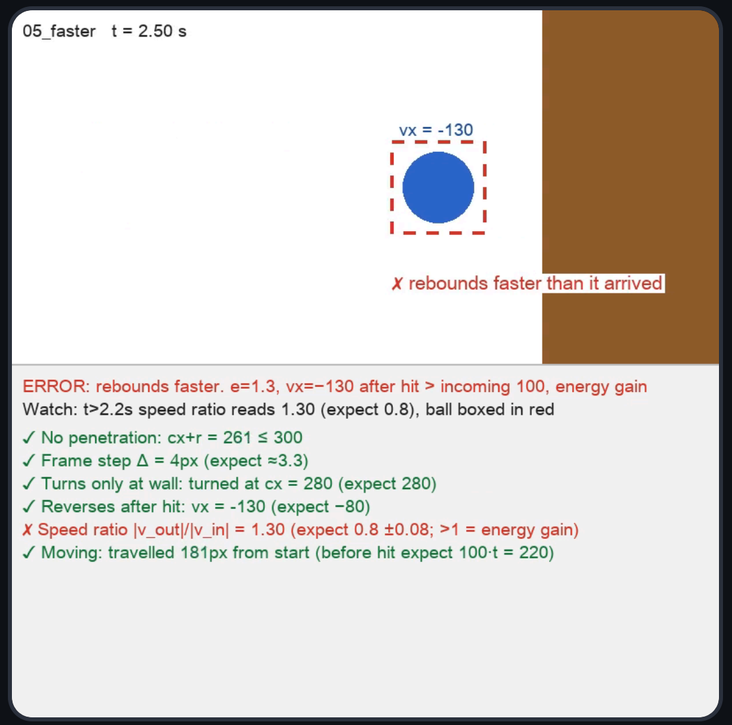


ERROR: turns early. Ball reverses at cx=220 (surface 60px from the wall) without contact
Watch: t=1.60s turn-around, red box shows a clear gap between ball and wall

✓ No penetration: cx+r = 168 ≤ 300
✓ Frame step Δ = 3px (expect ≈3.3)
✗ Turns only at wall: turned at cx = 220 (expect 280)
✓ Reverses after hit: vx = -80 (expect −80)
✓ Speed ratio |v_out|/|v_in| = 0.80 (expect 0.8 ±0.08; >1 = energy gain)
✓ Moving: travelled 88px from start (before hit expect 100·t = 220)

Turns back early

→ must reach the wall



ERROR: rebounds faster. e=1.3, vx=−130 after hit > incoming 100, energy gain
Watch: t>2.2s speed ratio reads 1.30 (expect 0.8), ball boxed in red

✓ No penetration: cx+r = 261 ≤ 300
✓ Frame step Δ = 4px (expect ≈3.3)
✓ Turns only at wall: turned at cx = 280 (expect 280)
✓ Reverses after hit: vx = -130 (expect −80)
✗ Speed ratio |v_out|/|v_in| = 1.30 (expect 0.8 ±0.08; >1 = energy gain)
✓ Moving: travelled 181px from start (before hit expect 100·t = 220)

Bounces back faster

→ energy gain is penalized

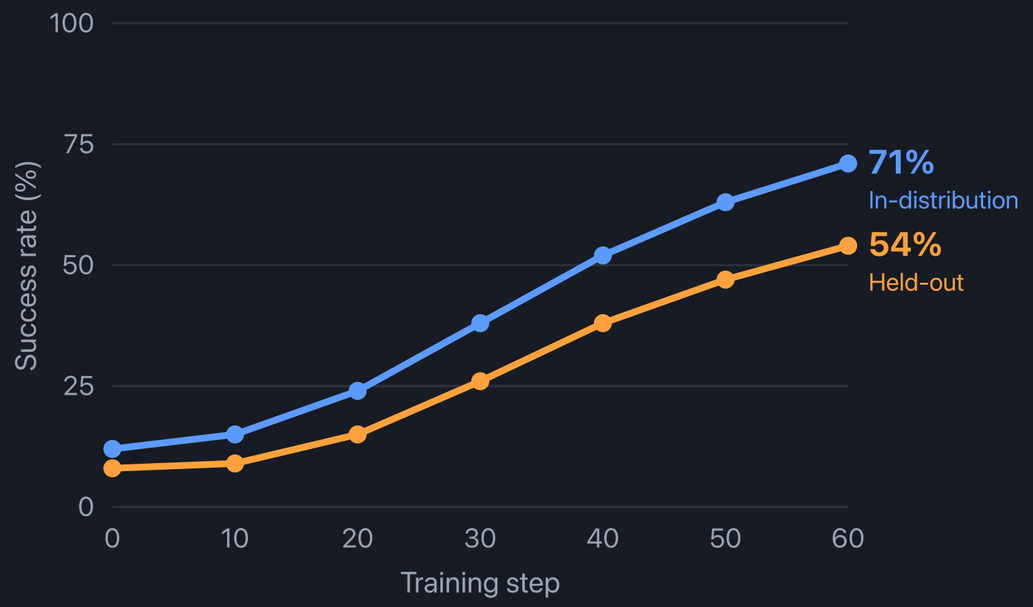
RESULTS

Success rate on cases it never saw.

Held-out scenes, before → after training

8% → 54%

	IN-DISTRIBUTION	HELD-OUT
Base model	12%	8%
After training	71%	54%



Success: no physics rule broken, and contact time, bounce speed, and trajectory all accurate. 200 scenes per set.